FinJeHeW Finnish Journal of eHealth and eWelfare

SCIENTIFIC PAPERS
VERTAISARVIOITU
KOLLEGIALT GRANSKAD
PEER-REVIEWED
www.tsv.fi/tunnus

Enabling the multiple sclerosis patient journey through digital technologies: Steps towards a digital care pathway in Finland

Jani Fält[1]*, Benjamin Paloniemi[1]*, Mervi Ryytty[1,2,3], Johanna Krüger[1,2,3], Guido Giunti[4,5,6]

[1] Neurology, Research Unit of Clinical Medicine, Faculty of Medicine, University of Oulu, Oulu, Finland; [2] Neurocenter, Neurology, Oulu University Hospital, Oulu, Finland; [3] MRC, Oulu University Hospital, Oulu, Finland; [4] Research Unit of Health Sciences and Technology, Faculty of Medicine, University of Oulu, Oulu, Finland; [5] Clinical Medicine Neurology, School of Medicine, Faculty of Health Sciences, Trinity College Dublin, Dublin, Ireland; [6] St. James's Hospital, Dublin, Ireland

*Jani Fält and Benjamin Paloniemi share the first authorship.

Guido Giunti, Research Unit of Health Sciences and Technology, Faculty of Medicine, P.O. BOX 5000, FI-90014, University of Oulu, FINLAND. Email: guido.giunti@oulu.fi

Abstract

Multiple sclerosis (MS) is a prevalent neurological disorder worldwide, leading to significant disability and impacting the quality of life and healthcare costs. The complex nature of MS necessitates a multidisciplinary approach involving various healthcare professionals to address its physical, cognitive, and psychological symptoms. Patient-centered care and digital technologies are becoming more relevant and embedded in disease management. Patient journeys are useful to understand pain points and opportunities for digital health interventions. This study aims to explore the journey of people with MS (pwMS) from the perspective of healthcare professionals in Finland and provide insights for digital health developers. A mixed methods approach was employed, conducting individual interviews with healthcare professionals and a narrative review of the current state of MS services. Thematic analysis of the interviews identified key stages of the patient journey and potential digital health touchpoints. The Finnish healthcare system offers various support services for MS patients, including medical, social, psychological, and rehabilitation aspects. Digital health technologies have the potential to enhance MS care at different stages of the patient journey. This study provides valuable insights into the MS patient journey and identifies opportunities for digital health interventions in Finland. Further research is warranted to validate and expand these findings to other settings and involve the perspectives of pwMS themselves.

Keywords: mHealth, digital health, multiple sclerosis, telemedicine, patient care management

Published under a CC BY 4.0 license (https://creativecommons.org/licenses/by/4.0/).

Introduction

Multiple sclerosis (MS) is one of the world's most common neurological disorders and is often the primary reason for non-traumatic neurological disability among young adults [1]. The number of people with MS (pwMS) in the world has increased from 2.3 million in 2013 to 2.8 million in 2020 and 2.9 million in 2023 [1]. More than 12 thousand Finns live with this condition. This chronic and progressive disease results in physical, cognitive, and psychological symptoms, leading to significant disability and impacting both the quality of life and healthcare costs [2,3]. The complex nature of MS demands a multidisciplinary approach to provide comprehensive care [4,5]. Efforts from healthcare professionals (HCPs) with diverse expertise, including neurologists, nurses, neuropsychologists, clinical psychologists, physiotherapists, occupational therapists, general practitioners, and other specialists, are needed to manage this condition [4–6].

Healthcare has been transitioning towards patient-centered care with the inclusion of patients in medical decision-making [7]. Including patients' perspectives has led to a transformation in the assessment of health outcomes [8,9]. User-centered design (UCD) is an approach to place the needs and characteristics of intended users in the center of software design and development [10,11]. User journey maps can act as a foundation for developing innovative solutions that solve problems and improve the overall healthcare experience for patients [12]. Mapping the journey of patients through their interactions with the healthcare system can provide valuable information [12–14]. Aspects of the patient experience may change when digital technologies are included in the care pathway [15].

The World Health Organization uses "digital health" as an umbrella term to encompass the field of knowledge and practice associated with the development and use of digital technologies to improve health [16]. This concept is used to include: eHealth, artificial intelligence, big data, blockchain, health data, health information systems, the infodemic, the Internet of Things, interoperability, mHealth, mobile apps and telemedicine. The use of digital health for pwMS has advanced rapidly [17]. There are mobile apps for PwMS for symptom tracking, medication reminders, lifestyle management, and access to educational resources [18,19]. There are digital health solutions for HCPs treating MS that enable clinical disability assessment, medical data analysis, and remote monitoring [20–22]. The use of wearable technologies to track health metrics, such as activity levels, sleep patterns, and heart rate can provide insights into the overall well-being of PwMS [17,23].

Worldwide, there are several registries and databases concerning MS, such as MSBase [24], gathering information from PwMS worldwide; NARCOMS, operating in North America; and the MS Register in Sweden and the United Kingdom in Europe [25]. In Finland, there is also a national MS registry (Neurorekisteri) that collects and stores information about individuals with MS. The Finnish MS registry also includes a patient portal called MyMS, where patients can input information that transfers to the view of healthcare professionals.

Care pathways descriptions have been developed for several conditions in Finland [26–28]; some have been published in Terveysportti, a health portal for HCPs [29]. Similarly, Current Care database (Käypä hoito) is a Finnish system that produces and maintains healthcare recommendations and evidence-based clinical practice guidelines, MS specific guidelines are among the currently available [30]. There are also Health Villages (Terveyskylä) [27], a Finnish online virtual health center, that

offer a broad spectrum of health-related information and support. Within this platform, the MS hub provides information specifically tailored for individuals with MS and their families, providing comprehensive resources, self-care programs, peer support, and expert guidance to manage life with MS; as well as information and tools for professionals.

However, there is no standardized patient journey that outlines potential digital touchpoints for MS. This leaves digital health designers and developers without clear guidance on potential opportunities for solutions for the various stages of MS care. This study aims to provide an exploratory view of the MS patient journey in Finland from the perspective of HCPs and provide a guide for digital health developers, identifying at which stages and how can the MS care pathway be supported with the help of digital health.

Methods

A mixed methods approach was taken in the present study. In order to understand the characteristics of HCPs who work with PwMS, a qualitative exploration through individual interviews was carried out. Purposive sampling was used to invite HCPs from the Neurology department of the Oulu University Hospital (OUH). In parallel, a narrative review of gray literature was conducted including websites, digital portals, and publicly available documents from the public sector or the third sector in Finland by examining their websites.

Setting

OUH is a healthcare provider, serving Northern Finland, which is home to 741,000 people and accounts for more than half of Finland's geographical area. The hospital offers specialized healthcare services to close to 1000 pwMS in the region.

Recruitment

HCPs from the Neurology department at OUH were recruited to the study. An email invitation was sent to HCPs in the department who directly treat or manage PwMS. The main inclusion criterion was direct experience in MS care and willingness to participate.

Exclusion criteria included professionals who were on extended leave or had no direct contact with PwMS. Participants were selected to account for the major occupations working with PwMS at OUH, resulting in a diverse sample of neurologists, nurses, psychologists, and neuropsychologists. Participants were coded as HCP from 01 to 08, e.g., HCP01.

Data collection

A semi-structured interview guide was composed ad-hoc using relevant items from the literature and UCD techniques. It included questions regarding the overall care pathway of MS and how digital health could be used to assist HCP in improving care. A video of a prototype gamified fatigue management digital health solution for MS [31], was used as a concrete case to trigger responses from HCPs. Emerging topics were explored with subsequent participants. All interviews were audio-recorded and transcribed verbatim, resulting in a total of 55 pages of raw transcript data.

The narrative review of the gray literature followed a search strategy that utilized resources from the Finnish Social Insurance Institution (Kela), Finnish Neuro Society (Neuroliitto) and its branch associations, Healthcare provider websites (e.g., Terveyskylä, Terveysportti), Private rehabilitation centers offering MS-related services. The search period occurred between January and June of 2023, reviewing the resources looking for mentions of

"Multiple Sclerosis," "chronic treatment," "digital health services," "rehabilitation," "patient support," "Kela," "Neuroliitto". Findings were included if they described services for MS, financial aids, rehabilitation courses, or digital tools accessible to Finnish residents. Exclusion criteria consisted of services with no presence in Finland.

Researcher reflexivity

The research team included experts in digital health and MS care. Researchers were aware that their interest in user-centered solutions could influence data collection (e.g., follow-up questions focusing on digital health needs). A reflexive approach was maintained by documenting assumptions prior to interviews, and by having multiple team members independently review transcripts to mitigate individual bias.

Data analysis

Interviews were conducted by native speakers in Finnish and coded using thematic analysis [32]. Thematic analysis was data-driven, as themes were derived directly from the content of the interviews rather than being pre-determined by existing theories. Two authors independently iteratively reviewed the transcripts, identifying recurring themes. Redundant themes were merged and removed involving a third author to achieve consensus.

The themes, stages and concepts were used to create a journey map [12,14] that visualized the interactions between pwMS and HCPs across different channels over time, from the HCPs perspective. Key digital touchpoints for MS were derived from recurring themes, specifically where HCPs mentioned opportunities for digital intervention or existing digital tools.

These narrative review findings are summarized separately from the interview findings to keep data sources clear.

Ethical considerations

According to the ethical principles of research with human participants and ethical review in the human sciences in Finland [33], ethical approval was not needed for the present study as it involved HCPs in their capacity as experts. Nevertheless, participants were given informed consent including their rights as study participants.

Results

A total of 8 interviews took place in the Neurology department with people who work with PwMS. In Table 1 and Table 2, we present some descriptive characteristics of the participants.

Familiarity with digital tools was quite high. The bulk of their digital activity seemed to be towards collaborating with other HCPs with very few exceptions regarding patients.

The majority had used apps and had recommended them to patients. HCPs had used apps also for education and networking purposes, followed by chronic disease management. A detailed view is provided in Figure 1.

Table 1. Participant characteristics.

Characteristics	Description
Gender	The majority of the HCPs were female, with 87.5% (7 out of 8) being women.
Age	The median age of HCPs was 54 years, with an interquartile range (IQR) of 19.5 years. This indicates that the group has a relatively wide age distribution, with a mix of mid-career and senior professionals.
Healthcare Profession	The sample included a diverse range of specialties: 25% neurologists (2 out of 8), 37.5% neuropsychologists (3 out of 8), and 37.5% nurses (3 out of 8). This variety highlights a multidisciplinary approach to healthcare within the sample.
Years of HCP experience	HCPs had a median of 21.5 years of experience in healthcare, with an IQR of 13.5 years. This suggests that the group is highly experienced, with a significant range of years in practice, potentially leading to a depth of expertise and knowledge.
Years of MS experience	On average, HCPs had been working with PwMS for 18 years, with an IQR of 18.5 years. This long-term experience with PwMS reflects a familiarity with the condition.

Table 2. Technology user characteristics.

Technology use	n (%)	
Smartphone users	8 (100.0%)	
Smartwatch users	2 (25.0%)	
Activity tracker users	5 (62.5%)	
Frequency of Mobile Phone Purchases/Upgrades		
Every 1 or 2 years	1 (12.5%)	
Between 2 and 3 years	1 (12.5%)	
More than 3 years	3 (37.5%)	
When needed (breaks or stops working)	2 (25.0%)	
Experience with technology	Use with Patients	Use with Colleagues
Sharing files (Dropbox, Box, Amazon Cloud Drive, etc)	0 (0.0%)	2 (25.0%)
Social media interactions (Facebook, Instagram, Twitter, etc)	0 (0.0%)	3 (37.5%)
Instant messaging (Messenger, WhatsApp, Telegram, etc)	1 (12.5%)	4 (50.0%)
Office suites (Google Drive, Office365, etc)	2 (25.0%)	5 (62.5%)
Recommended the use of digital health apps to patients	6 (75.0%)	

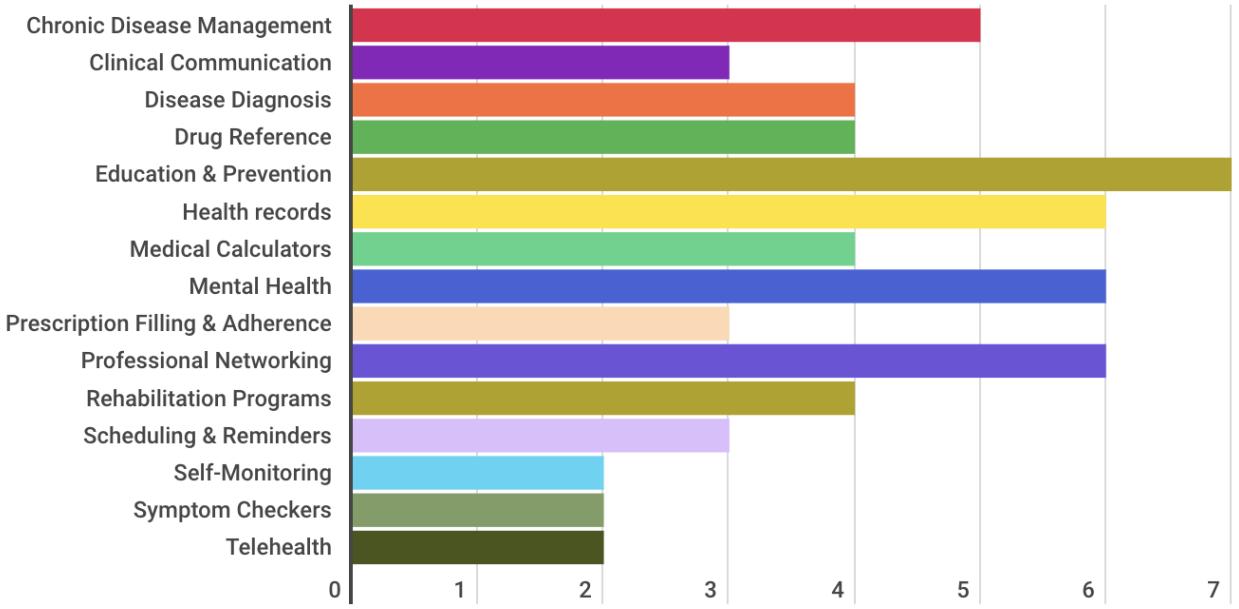


Figure 1. mHealth apps used by healthcare professionals.

A Narrative review on MS support services

An overview of the services available to pwMS in Finland is provided. Table 3 presents the sources included in the narrative review, indicating the type of resource, and issuing organization. It also describes how these references informed our understanding of existing support systems and digital solutions for MS in Finland.

The Finnish healthcare system offers a range of services and support systems for pwMS. Public healthcare providers offer medical care for PwMS, including neurologists, specialized nurses, and other HCPs.

Finland has strong legislation promoting accessibility for people with disabilities. It is part of Finnish law that the wellbeing service counties must provide services to people with disabilities [34,35]. Public transportation, buildings, and other facilities are designed to accommodate individuals with mobility impairments, making it easier for those with MS to maintain independence and participate in society

Kela supports pwMS in different aspects. From a financial perspective, it offers various financial aid to individuals with disabilities, including those with MS, like disability pensions, rehabilitation allowances, and others [36]. Kela also offers adjustment training courses for newly diagnosed where the person with MS (and their family members) can go through group conversations, workshops and meetings with different HCPs [37]. Access to mental health support is available to help cope with the emotional aspects of MS.

Public and private rehabilitation centers offer services such as physiotherapy, occupational therapy, and speech therapy to help patients manage their symptoms and maintain independence. Kela provides a rehabilitation course that aims to provide support and guidance in everyday life [38].

Various organizations and support groups exist, such as Neuroliitto, which is a community of 25

member associations for rare neurological diseases. These groups connect patients, provide educational resources, and advocate for better care and support services [39]. The Neuroliitto branch in Pirkanmaa has developed a map of services for pwMS [40], represented in Figure 2 in English.

The Finnish MS register available for public health care organizations is provided by software company StellarQ (stellarq.com). In 2017 a patient reported outcomes tool was added to the system [41], and already in 2018 [42], 15 of the 21 of Finland's public hospital districts have joined the service, and most of them have integrated it into their electronic patient documentation system.

Table 3. Sources Used for the Narrative Review and the reason for their inclusion.

Source / Reference	Type	Reason for Inclusion
Rauhala M, Kinnunen U-M. [27] Terveyskylässä palvelua asiakkaalle.	Journal Article	Description of Terveyskylä services and how they benefit patients in Finland, providing background on digital health tools.
Duodecim. Terveysportti 20 vuotta – mistä kaikki alkoi? [29]	Web article	Describes the history and role of Terveysportti, which publishes care pathways (hoitoketjut) for many conditions, including MS, used by Finnish HCPs.
[Suomalaisen Lääkäriseuran Duodecimin ja Suomen Neurologinen Yhdistys ry:n asettama työryhmä. [30]	Clinical Guideline	Official guideline outlining evidence-based MS diagnosis, treatment, and follow-up in Finland; essential for understanding standard MS care pathways.
Ministry of Social Affairs and Health. Disability Services Act. 675/2023, Finland [34]	Government Legislation	Provides legal framework ensuring individuals with disabilities (including advanced MS) receive adequate services and support from Finnish wellbeing service counties.
Ministry of Social Affairs and Health. Act on services and support measures organized on the basis of disability. 380/1987, Finland[35]	Government Legislation	Establishes legal provisions for disability services in Finland, relevant for PwMS requiring mobility or rehabilitation support.
Kela. Lääkkeet ja lääkekorvaukset, MS-tauti [36]	Social Insurance Institution Webpage	Outlines pharmaceutical reimbursements and coverage for MS medications, illustrating financial support mechanisms within the Finnish system.
Kela. Sopeutumisvalmennuskurssi MS- tautia sairastavalle [37]	Social Insurance Institution Webpage	Describes adaptation training courses offered for newly diagnosed PwMS and their families in Finland, showing rehabilitation options.
Kela. Kuntoutuskurssi MS-tautia sairastavalle [38]	Social Insurance Institution Webpage	Provides details on rehabilitation courses specifically for PwMS, supporting the narrative of available public services in Finland.
Neuroliitto. MS-tauti vasta-sairastuneelle [39].	Patient Association Webpage	A comprehensive handbook for newly diagnosed PwMS; demonstrates the type of educational materials provided by patient organizations in Finland.
Pirkanmaan neuroyhdistys. Palvelukartta [40]	Patient Association Webpage	Illustrates the service map for PwMS in the Pirkanmaa region; used in the manuscript as an example of multi-sector resource mapping.
Rantamäki M. MS-potilaan hoito ja kuntoutus Etelä-Pohjanmaan sairaanhoitopiirissä [43]	Hospital Webpage	One of the earlier documents mapping MS care and rehabilitation at the hospital district level in Finland, providing historical or contextual background on MS patient pathways.



Figure 2. MS related services in Finland, adapted from Neuroliitto branch in Pirkanmaa.

MS patient journey

Interviews with the different HCPs were analyzed thematically and synthetized with the results from the narrative review to construct the different stages of the disease journey. Quotes from the HCPs are placed in Table 4 according to the different stages.

The different perceived needs led to identified opportunities for digital health solutions shown in Table 5 as a list of desired features per stage of the patient journey.

Table 4. Illustrative MS patient journey according to HCP quotes.

PROFILE	PRE-DIAGNOSIS Symptoms raise concern leading to evaluations	DIAGNOSIS	TREATMENT Medication Lifestyle changes Rehabilitation	RELAPSE
Person with MS		HCP04: The person needs to understand the disease	HCP04: The patient develops their own personal relationship with the disease HCP04: The family needs to understand the disease	
General practitioner	HCP02: Symptoms start to affect life		HCP01: The patient needs to learn new lifestyles	HCP01: The patient needs to learn new lifestyles
Nurse	HCP01: Symptoms disturb life	HCP05: Diagnosis is a shock for patients	HCP05: Treatment starts with an MS-nurse HCP03: Fatigue disturbs life	HCP01: Fatigue becomes too difficult to control
Psychologist Neuro -psychologist	HCP03: There may appear cognitive symptoms		HCP07: MS-patients come to a neurologists appointment. HCP03: Fatigue can cause mental health symptoms HCP06: Rehabilitation supports fatigue management	HCP08: The role of rehabilitation increases as the disease progresses HCP03: Fatigue can cause mental health symptoms
Neurologist		HCP07: We are responsible for the diagnosis and treatment	HCP07: We answer consultations about MS-patients HCP07: Fatigue affects compliance to medication HCP03: The patient may need a neuropsychologist	

Table 5. Digital health opportunities of the MS patient journey according to HCPs.

PRE-DIAGNOSIS Symptoms raise concern leading to evaluations	DIAGNOSIS	TREATMENT Medication Lifestyle changes Rehabilitation	RELAPSE
Symptom checker Interactive disease path	Patient education Exam results Coaching	Medication management Symptom tracker Family support Community of peers Fatigue management Personalized recommendations Online resources Physical activity tracking Reminders Cognitive stimulation Managing work-related stress	Relapse detection

Discussion

We conducted interviews with healthcare professionals at the OUH. Participants were interviewed from all the main occupations who are in contact with PwMS on a daily basis at the OUH. Our work provides insights into how PwMS are treated in Finland, what their journey as patients through the healthcare system looks like and what support systems and services are in place for PwMS in Finland. We also identified points in the patient journey where digital health technology could be used to aid in the treatment of MS disease.

While our participant sample is small and specific to a single center, the qualitative insights shed light on common challenges and opportunities for digital health in MS care from the perspective of HCPs. Our goal is not statistical generalization but rather a richer understanding of this context.

Despite Finland's resources (Käypä hoito guidelines, Neurorekisteri with MyMS, and Terveyskylä's MS-talo), our findings indicate that HCPs still see opportunities to expand digital health, especially in areas such as personalized self-monitoring, seamless data sharing across care levels, and patient education. This is in line with the literature, as it has been pwMS and their caregivers are receptive to, and see a need for digital health for MS [20,44]. Similar sentiments towards the development of digital health tools from the perspective of patients were found by Madanian et al. [45].

To our knowledge this is the first scientific publication exploring the patient journey of PwMS and its touchpoints with HCPs in Finland. In 2001 the Hospital District of South Ostrobothnia mapped the journey for MS regarding treatment and rehabilitation [43], but such work can be considered outdated due to the recent changes in the Finnish healthcare system. A few hospital districts have developed their own care pathways for MS disease, but these have not been made public in peer-reviewed forums.

The need for a unifying patient portal where HCPs can document information and where patients can access their health data seemed to be present. A unifying patient portal may also have many positive impacts and potential for pwMS and their caregivers [44, 46]. In Finland there are many different patient record systems in use as well as a nationwide patient portal OmaKanta.

To our knowledge this is also the first publication to explore opportunities for the development of digital health via a patient journey. It is important to understand the patient journey because it allows healthcare providers to improve the patient experience and make informed decisions [47]. Understanding the patient journey also reveals effective points of intervention for digital health and allows us to design more effective digital health [46].

Digital health is changing the way MS care is provided by increasing patients' engagement [48]. These technologies have been shown to be effective at delivering care to PwMS [23], and their use is expected to grow. A recent publication on the MyMS an interface from the Finnish MS Register reported that said interface is used by 10% of Finnish individuals with MS [41]. As the goal of this solution is to improve patient-centeredness and autonomy, the use of UCD could help in designing solutions that are more specific to the needs of the intended users with higher acceptance and lower user errors [49–51]. Involving public and patient representatives in the design process may provide valuable insights [52,53].

Limitations

This study has limitations that are inherent to qualitative research methods. The sample size (N=8 HCPs) is not representative of all HCPs who manage MS across Finland, and our study focused in a single center. Other professional groups were not included. The findings are therefore context-specific and not meant for broad generalization.

As participants come from a highly developed country such as Finland, it may reduce the scalability of the results, as the quality of care, healthcare system organization, economic status, etc. differs from other countries.

Further, It should be noted that prompting HCPs with the video of a fatigue management solution may have biased responses towards fatigue. Finally, as this work is based on the experience of HCPs, it may not reflect the experiences of pwMS. In addition, the research team's background and interest in digital health could have shaped probing questions and interpretations during analysis.

The narrative review of gray literature offered only an illustrative rather than exhaustive overview of MS-related services in Finland. A more comprehensive systematic review might reveal additional or different resources and digital solutions.

Future work

Future research should incorporate patient perspectives on their experiences, emotions, and expectations. A comprehensive review of the journey from primary to specialist care, along with existing support solutions in Finland, would be valuable. Comparing HCP and patient perspectives across regions and countries could provide deeper insights into MS care variations. Broader evaluations may reveal new approaches. As digitalization advances in Finnish healthcare, identifying key points for digital interventions through patient journey analysis can aid in developing effective solutions.

Contributions

JF and BP contributed equally and are listed as co-first authors. JF, BP and GG conceived, designed, led overall study conduct. JF and BP carried out data collection. JF, BP and GG carried out analysis and interpretation of the data. MR was involved in the design of the interview guide and study permissions. MR and JK were involved in the participant recruitment, analysis and interpretation of study data. All authors contributed to the manuscript drafting process, read and approved the final manuscript.

Acknowledgements

The authors would like to thank study nurse Pirkko Holappa at the Oulu University Hospital for her excellent work in helping to recruit patients for this study. We also thank the participating HCPs.

GG would like to thank Prof. Minna Isomursu, Vasiliki Mylonopoulou PhD, Prof. Octavio Rivera-Romero, Tiia Yrttiaho, Sharon Guardado, and Irantzu Garces Munarriz for their collaboration and support.

The present study was partly funded by the European Union's Horizon 2020 Research and Innovation Programme under the Marie Skłodowska-Curie grant agreement No. 101034252. The publication has also emanated from research supported (in part) by a research grant from Science Foundation Ireland (SFI) under Grant Number 16/RC/3948, and Business Finland's More Stamina Research to Business project.

Conflict of interests

None declared.

Abbreviations

BP: Benjamin Paloniemi
GG: Guido Giunti
HCP: Healthcare Professional
IQR: Interquartile range
JF: Jani Fält
JK: Johanna Krüger
Kela: Finnish Social Insurance Institution
mHealth: mobile health
MR: Mervi Ryytty
MS: Multiple Sclerosis
Neuroliitto: Finnish Neuro Society
OUH: Oulu University Hospital
pwMS: people with MS.

References

[1] Browne P, Chandraratna D, Angood C, Tremlett H, Baker C, Taylor BV, Thompson AJ. Atlas of Multiple Sclerosis 2013: A growing global problem with widespread inequity. Neurology. 2014 Sep 9;83(11):1022-4. https://doi.org/10.1212/WNL.0000000000000768

[2] Kobelt G, Thompson A, Berg J, Gannedahl M, Eriksson J; MSCOI Study Group; European Multiple Sclerosis Platform. New insights into the burden and costs of multiple sclerosis in Europe. Mult Scler. 2017 Jul;23(8):1123-1136. https://doi.org/10.1177/1352458517694432

[3] Kobelt G, Berg J, Lindgren P, Fredrikson S, Jönsson B. Costs and quality of life of patients with multiple sclerosis in Europe. J Neurol Neurosurg Psychiatry. 2006 Aug;77(8):918-26. https://doi.org/10.1136/jnnp.2006.090365

[4] Soelberg Sorensen P, Giovannoni G, Montalban X, Thalheim C, Zaratin P, Comi G. The Multiple Sclerosis Care Unit. Mult Scler. 2019 Apr;25(5):627-636. https://doi.org/10.1177/1352458518807082

[5] Savio M, Kearney H, Giunti G. Evaluating the evidence behind multidisciplinary roles for a multiple sclerosis unit: A systematic literature review. Mult Scler Relat Disord. 2025 Mar;95:106342. https://doi.org/10.1016/j.msard.2025.106342

[6] Papeix C, Gambotti L, Assouad R, Ewenczyck C, Tanguy ML, Pineau F, Houis MC, Mazevet D, Maillart E, Lubetzki C. Evaluation of an integrated multidisciplinary approach in multiple sclerosis care: A prospective, randomized, controlled study. Mult

Scler J Exp Transl Clin. 2015 Sep 28;1:2055217315608864. https://doi.org/10.1177/2055217315608864

[7] Doyle C, Lennox L, Bell D. A systematic review of evidence on the links between patient experience and clinical safety and effectiveness. BMJ Open. 2013 Jan 3;3(1):e001570. https://doi.org/10.1136/bmjopen-2012-001570

[8] D'Amico E, Haase R, Ziemssen T. Review: Patient-reported outcomes in multiple sclerosis care. Mult Scler Relat Disord. 2019 Aug;33:61-66. https://doi.org/10.1016/j.msard.2019.05.019

[9] Gil-Perotin S, Bernad L, Reddam S, Ferrer-Pardo C, Navarro-Quevedo S, Solís-Tarazona L. Patient's perspective in clinical practice to assess and predict disability in multiple sclerosis. Sci Rep. 2022 Oct 29;12(1):18238. https://doi.org/10.1038/s41598-022-23088-x

[10] Giunti G, Kool J, Rivera Romero O, Dorronzoro Zubiete E. Exploring the Specific Needs of Persons with Multiple Sclerosis for mHealth Solutions for Physical Activity: Mixed-Methods Study. JMIR Mhealth Uhealth. 2018 Feb 9;6(2):e37. https://doi.org/10.2196/mhealth.8996

[11] Peng W, Yuan S, Holtz BE. Exploring the Challenges and Opportunities of Health Mobile Apps for Individuals with Type 2 Diabetes Living in Rural Communities. Telemed J E Health. 2016 Sep;22(9):733-8. https://doi.org/10.1089/tmj.2015.0180

[12] Trebble TM, Hansi N, Hydes T, Smith MA, Baker M. Process mapping the patient journey: an introduction. BMJ. 2010 Aug 13;341:c4078. https://doi.org/10.1136/bmj.c4078

[13] Lamprell K, Braithwaite J. Patients as story-tellers of healthcare journeys. Med Humanit. 2016 Sep;42(3):207-9. https://doi.org/10.1136/medhum-2016-010885

[14] Simonse L, Albayrak A, Starre S. Patient journey method for integrated service design. Design for Health 2019;3(1):82–97. https://doi.org/10.1080/24735132.2019.1582741

[15] Wang T, Giunti G, Melles M, Goossens R. Digital Patient Experience: Umbrella Systematic Review. J Med Internet Res. 2022 Aug 4;24(8):e37952. https://doi.org/10.2196/37952

[16] World Health Organization. EUR/RC72/5: Regional digital health action plan for the WHO European Region 2023–2030. World Health Organization; 2022. https://iris.who.int/handle/10665/360950

[17] Ziemssen T, Haase R. Digital Innovation in Multiple Sclerosis Management. Brain Sci. 2021 Dec 29;12(1):40. https://doi.org/10.3390/brainsci12010040

[18] Giunti G, Guisado Fernández E, Dorronzoro Zubiete E, Rivera Romero O. Supply and Demand in mHealth Apps for Persons With Multiple Sclerosis: Systematic Search in App Stores and Scoping Literature Review. JMIR Mhealth Uhealth. 2018 May 23;6(5):e10512. https://doi.org/10.2196/10512

[19] Giunti G, Yrttiaho T, Guardado-Medina S, Sachinopoulou A, Mylonopoulou V, Fält J, Paloniemi B, Ryytty M, Krüger J, Isomursu M. Feasibility and usability evaluation of a gamified fatigue management mobile application for persons with multiple sclerosis in everyday life. Mult Scler Relat Disord. 2025 May;97:106379. https://doi.org/10.1016/j.msard.2025.106379

[20] Guardado S, Isomursu M, Giunti G. Health Care Professionals' Perspectives on the Uses of Patient-Generated Health Data. Stud Health Technol Inform. 2022 May 25;294:750-754. https://doi.org/10.3233/SHTI220577

[21] Gagnon MP, Ngangue P, Payne-Gagnon J, Desmartis M. m-Health adoption by healthcare

professionals: a systematic review. J Am Med Inform Assoc. 2016 Jan;23(1):212-20. https://doi.org/10.1093/jamia/ocv052

[22] Lluch M. Healthcare professionals' organisational barriers to health information technologies-a literature review. Int J Med Inform. 2011 Dec;80(12):849-62. https://doi.org/10.1016/j.ijmedinf.2011.09.005

[23] Yeroushalmi S, Maloni H, Costello K, Wallin MT. Telemedicine and multiple sclerosis: A comprehensive literature review. J Telemed Telecare. 2020 Aug-Sep;26(7-8):400-413. https://doi.org/10.1177/1357633X19840097

[24] Kalincik T, Butzkueven H. The MSBase registry: Informing clinical practice. Mult Scler. 2019 Dec;25(14):1828-1834. https://doi.org/10.1177/1352458519848965

[25] Marrie RA, Cutter GR, Fox RJ, Vollmer T, Tyry T, Salter A. NARCOMS and Other Registries in Multiple Sclerosis: Issues and Insights. Int J MS Care. 2021 Nov-Dec;23(6):276-284. https://doi.org/10.7224/1537-2073.2020-133

[26] Ortiz F, Giunti G. Usability assessment of an interactive health technology for kidney living donors: protocol for a prospective cross-sectional survey. BMJ Open. 2022 Jan 3;12(1):e051166. https://doi.org/10.1136/bmjopen-2021-051166

[27] Rauhala M, Kinnunen UM. Terveyskylässä palvelua asiakkaalle. FinJeHeW 2017;9(2-3):251-258. https://doi.org/10.23996/fjhw.60948

[28] Ortiz F, Grasberger J, Ekstrand A, Helanterä I, Giunti G. Interactive Health Technology Tool for Kidney Living Donor Assessment to Standardize the Informed Consent Process: Usability and Qualitative Content Analysis. JMIR Form Res. 2024 Jul 9;8:e47785. https://doi.org/10.2196/47785

[29] Duodecim. Terveysportti 20 vuotta – mistä kaikki alkoi? Duodecim; 2020 [cited 20 March 2024] Available from: https://www.duodecim.fi/2020/12/18/terveysportti-20-vuotta-mista-kaikki-alkoi/

[30] Suomalaisen Lääkäriseuran Duodecimin ja Suomen Neurologinen Yhdistys ry:n asettama työryhmä. MS-tauti. Käypä hoito. Suomalainen Lääkäriseura Duodecim; 2024 [cited 20 March 2024]. Available from: https://www.kaypahoito.fi/hoi36070

[31] Giunti G, Mylonopoulou V, Rivera Romero O. More Stamina, a Gamified mHealth Solution for Persons with Multiple Sclerosis: Research Through Design. JMIR Mhealth Uhealth. 2018 Mar 2;6(3):e51. https://doi.org/10.2196/mhealth.9437

[32] Braun V, Clarke V. Using thematic analysis in psychology. Qual Res Psychol 2006;3(2):77–101. https://doi.org/10.1191/1478088706qp063oa

[33] Finnish Advisory Board on Research Ethics. Ethical principles of research in the humanities and social and behavioural sciences and proposals for ethical review. National Advisory Board on Research Ethics; 2009. Available from: http://www.tenk.fi/sites/tenk.fi/files/ethicalprinciples.pdf

[34] Finlex. Vammaispalvelulaki [Disability Services Act] 675/2023. Ministry of Social Affairs and Health; 2023 [cited 12 Sept 2025]. Available from: https://www.finlex.fi/en/legislation/collection/2023/675

[35] Finlex. Laki vammaisuuden perusteella järjestettävistä palveluista ja tukitoimista [Act on services and support measures organized on the basis of disability] 380/1987. Ministry of Social Affairs and Health; 1987 [cited 12 Sept 2025]. Available from: https://www.finlex.fi/fi/lainsaadanto/1987/380

[36] Kela. Lääkkeet ja lääkekorvaukset, Korvausoikeudet, Erityiskorvaus, 109 MS-tauti. Kela; 2023

[cited 6 June 2023]. Available from: https://www.kela.fi/laake109

[37] Kela. Sopeutumisvalmennuskurssi MS-tautia sairastavalle. Kela; 2023 [cited 6 June 2023]. Available from: https://www.kela.fi/ms-tauti-sopeutumisvalmennus

[38] Kela. Kuntoutuskurssi MS-tautia sairastavalle. Kela; 2023 [cited 6 June 2023]. Available from: https://www.kela.fi/ms-tauti-kuntoutuskurssi

[39] Neuroliitto. MS-tauti käsikirja vastasairastuneelle. Raisio: Newprint Oy; 2020 [cited 12 Sept 2025]. Available from: https://neuroliitto.fi/tuote/ms-tauti-kasikirja-vasta-sairastuneelle/

[40] Pirkanmaan neuroyhdistys. Palvelukartta. Pirkanmaan neuroyhdistys ry; 2023 [cited 6 June 2023]. Available from: https://neuropi.fi/palvelukartta/

[41] Hämäläinen P, Viitala M, Kuusisto H, Ruutiainen J, Soilu-Hänninen M. MyMS: An Interface for Patient-Reported Outcomes for Finnish Individuals With Multiple Sclerosis. Int J MS Care. 2024 Oct 7;26(Q4):273-280. https://doi.org/10.7224/1537-2073.2023-082

[42] Laakso SM, Viitala M, Kuusisto H, Sarasoja T, Hartikainen P, Atula S, Tienari PJ, Soilu-Hänninen M. Multiple sclerosis in Finland 2018-Data from the national register. Acta Neurol Scand. 2019 Nov;140(5):303-311. https://doi.org/10.1111/ane.13145

[43] Rantamäki M. MS-potilaan hoito ja kuntoutus Etelä-Pohjanmaan sairaanhoitopiirissä. Seinäjoki: Etelä-Pohjanmaan sairaanhoitopiiri; 2001.

[44] Haase R, Voigt I, Scholz M, Schlieter H, Benedict M, Susky M, Dillenseger A, Ziemssen T. Profiles of eHealth Adoption in Persons with Multiple Sclerosis and Their Caregivers. Brain Sci. 2021 Aug 19;11(8):1087. https://doi.org/10.3390/brainsci11081087

[45] Madanian S, Nakarada-Kordic I, Reay S, Chetty T. Patients' perspectives on digital health tools. PEC Innov. 2023 May 26;2:100171. https://doi.org/10.1016/j.pecinn.2023.100171

[46] Voigt I, Benedict M, Susky M, Scheplitz T, Frankowitz S, Kern R, Müller O, Schlieter H, Ziemssen T. A Digital Patient Portal for Patients With Multiple Sclerosis. Front Neurol. 2020 May 22;11:400. https://doi.org/10.3389/fneur.2020.00400

[47] Gualandi R, Masella C, Viglione D, Tartaglini D. Exploring the hospital patient journey: What does the patient experience? PLoS One. 2019 Dec 5;14(12):e0224899. https://doi.org/10.1371/journal.pone.0224899

[48] Bevens W, Weiland T, Gray K, Jelinek G, Neate S, Simpson-Yap S. Attrition Within Digital Health Interventions for People With Multiple Sclerosis: Systematic Review and Meta-analysis. J Med Internet Res. 2022 Feb 9;24(2):e27735. https://doi.org/10.2196/27735

[49] De Vito Dabbs A, Myers BA, Mc Curry KR, Dunbar-Jacob J, Hawkins RP, Begey A, Dew MA. User-centered design and interactive health technologies for patients. Comput Inform Nurs. 2009 May-Jun;27(3):175-83. https://doi.org/10.1097/NCN.0b013e31819f7c7c

[50] Pruitt J, Adlin T. The Persona Lifecycle: Keeping People in Mind Throughout Product Design. San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.; 2005.

[51] Johnson CM, Johnson TR, Zhang J. A user-centered framework for redesigning health care interfaces. J Biomed Inform. 2005 Feb;38(1):75-87. https://doi.org/10.1016/j.jbi.2004.11.005

[52] Yrttiaho T, Giunti G, Isomursu M. Challenges Implementing Patient and Public Involvement in a

Digital Health Agile Project Which Includes Research, Business and Software Development. Stud Health Technol Inform. 2024 Aug 22;316:310-314. https://doi.org/10.3233/SHTI240406

[53] Yrttiaho T, Isomursu M, Giunti G. Experiences Using Patient and Public Involvement in Digital Health Research for Multiple Sclerosis. Stud Health Technol Inform. 2022 May 25;294:735-739. https://doi.org/10.3233/SHTI220574